

Full Solution Candidate: The Converse to Proposition 4.9 in *The approximation property and Lipschitz mappings on Banach spaces*

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Status

This is a candidate full solution, subject to human review, for the open converse stated on page 6 of Rueda–Sanchez-Perez [1]. It addresses only that Lipschitz-operator-ideal converse. It does not address the separate classical approximation-property problem on page 2 of the same paper.

Source Question

The source discusses the set

$$\mathcal{O}_0(X, \mathcal{A}\mathcal{E}(X)) = \text{Lip}_{0\mathcal{F}}(X, \mathcal{A}\mathcal{E}(X))$$

and states Proposition 4, quoted from Achour–Rueda–Sanchez-Perez–Yahi [2]: if a pointed metric space X has the $\mathcal{I}$ -Lipschitz approximation property with respect to $\text{Lip}_{0\mathcal{F}}(X, \mathcal{A}\mathcal{E}(X))$, then for every pointed metric space Z and every $\mathcal{I}$ -Lipschitz compact mapping $\phi : Z \rightarrow X$, the map $\delta_X \circ \phi$ can be approximated uniformly on B_Z by finite-rank Lipschitz maps in $\text{Lip}_{0\mathcal{F}}(Z, \mathcal{A}\mathcal{E}(X))$. Immediately before this proposition the source says that the converse is still open.

The set $\mathcal{O}_0(X, \mathcal{A}\mathcal{E}(X)) = \text{Lip}_{0\mathcal{F}}(X, \mathcal{A}\mathcal{E}(X))$, allows to get the approximation of vector-valued Lipschitz mapping by means of finite rank Lipschitz mappings.

Proposition 3. [1, Proposition 4.8] *Let X be a pointed metric space. The following assertions are equivalent:*

- X has the $\mathcal{I}$ -Lipschitz approximation property with respect to $\text{Lip}_{0\mathcal{F}}(X, \mathcal{A}\mathcal{E}(X))$.
- For every Banach space E , $\text{Lip}_{0\mathcal{F}}(X, E)$ is Lipschitz- $\tau_{\mathcal{I}}$ dense in $\text{Lip}_0(X, E)$.

It also permits to obtain the approximation of $\mathcal{I}$ -Lipschitz compact operators by finite rank Lipschitz mappings as shows the following proposition. However the converse is still an open problem.

Proposition 4. [1, Proposition 4.9] *Let X be a pointed metric space and $\mathcal{I}$ be an operator ideal. If X has the $\mathcal{I}$ -Lipschitz approximation property with respect to $\text{Lip}_{0\mathcal{F}}(X, \mathcal{A}\mathcal{E}(X))$ then, for any pointed metric space Z and any $\mathcal{I}$ -Lipschitz compact mapping $\phi : Z \rightarrow X$, the mapping $\delta_X \circ \phi$ can be approximated by finite rank operators of $\text{Lip}_{0\mathcal{F}}(Z, \mathcal{A}\mathcal{E}(X))$ uniformly on B_Z .*

When we consider the set $\mathcal{O}_1(E, \mathcal{A}\mathcal{E}(E)) = \delta_E \circ \text{Lip}_{0\mathcal{F}}(E, \mathcal{A}\mathcal{E}(E))$ for a Banach space E , then the $\mathcal{I}$ -Lipschitz approximation property is weaker than the $\mathcal{K}_{\mathcal{I}}$ -uniform approximation property.

Definitions and Formulation

Let X be a pointed metric space with base point 0, let $\text{AE}(X)$ be its Arens–Eells, equivalently Lipschitz-free, space, and let $\delta_X : X \rightarrow \text{AE}(X)$ be the Dirac map. A map belongs to $\text{Lip}_{0\mathcal{F}}(X, E)$ if it is base-point preserving, Lipschitz, and has finite-dimensional linear span of its range.

For an operator ideal $\mathcal{I}$, a subset $K \subset X$ is relatively $\mathcal{I}$ -Lipschitz compact when $\delta_X(K)$ is relatively $\mathcal{I}$ -compact in $\text{AE}(X)$. The $\mathcal{I}$ -Lipschitz approximation property with respect to $\text{Lip}_{0\mathcal{F}}(X, \text{AE}(X))$ means that δ_X lies in the topology of uniform convergence on all relatively $\mathcal{I}$ -Lipschitz compact subsets of X , after closure of $\text{Lip}_{0\mathcal{F}}(X, \text{AE}(X))$.

Proof Intuition

The converse has only one missing ingredient. To test the $\mathcal{I}$ -Lipschitz approximation property, one fixes a relatively $\mathcal{I}$ -Lipschitz compact set $K \subset X$ and must approximate δ_X on K by a finite-rank Lipschitz map defined on all of X . The assumed converse condition can be applied to the inclusion of $K \cup \{0\}$ into X , after rescaling the metric so that K is inside the unit ball. This gives the right finite-rank approximation on the small metric space. Because the range is finite-dimensional, extend the coordinate functions one at a time by McShane’s scalar extension theorem and then reassemble them. The extension remains finite rank and agrees with the local approximant on K .

Lemma 1 (Finite-rank Lipschitz extension). *Let A be a pointed subset of a metric space X , with the inherited metric and the same base point. If $f \in \text{Lip}_{0\mathcal{F}}(A, E)$, then there is $F \in \text{Lip}_{0\mathcal{F}}(X, E)$ such that $F|_A = f$.*

Proof. Let $Y = \text{span } f(A)$, which is finite-dimensional. Choose a basis $y_1, \dots, y_m$ of Y and coordinate functionals $\lambda_1, \dots, \lambda_m \in Y^*$. Each scalar function $\lambda_j \circ f$ is Lipschitz on A and vanishes at the base point. By the McShane extension theorem, in the real case, it extends to a scalar Lipschitz function $\tilde{\lambda}_j : X \rightarrow \mathbb{R}$ vanishing at the base point. In the complex case, apply the same real theorem to real and imaginary parts. Then

$$F(x) = \sum_{j=1}^m \tilde{\lambda}_j(x) y_j$$

is a base-point preserving Lipschitz map from X into the finite-dimensional space Y , and $F|_A = f$. $\square$

Theorem 1. *Let X be a pointed metric space and $\mathcal{I}$ an operator ideal. Assume that for every pointed metric space Z and every $\mathcal{I}$ -Lipschitz compact map $\phi : Z \rightarrow X$, the map $\delta_X \circ \phi$ can be approximated uniformly on B_Z by maps in $\text{Lip}_{0\mathcal{F}}(Z, \text{AE}(X))$. Then X has the $\mathcal{I}$ -Lipschitz approximation property with respect to $\text{Lip}_{0\mathcal{F}}(X, \text{AE}(X))$.*

Proof. Let $K \subset X$ be relatively $\mathcal{I}$ -Lipschitz compact and let $\varepsilon > 0$. We must find $G \in \text{Lip}_{0\mathcal{F}}(X, \text{AE}(X))$ such that

$$\sup_{x \in K} \|\delta_X(x) - G(x)\|_{\text{AE}(X)} < \varepsilon.$$

The set K is bounded, since $\delta_X(K)$ is relatively $\mathcal{I}$ -compact in the Banach space $\text{AE}(X)$. Put

$$R = \max\{1, \sup_{x \in K} d_X(x, 0)\}$$

and let $Z = K \cup \{0\}$, pointed at 0, with metric

$$d_Z(z, z') = R^{-1} d_X(z, z').$$

Then $Z = B_Z$. Let $\phi : Z \rightarrow X$ be the inclusion. It is Lipschitz, and

$$\delta_X(\phi(B_Z)) = \delta_X(K \cup \{0\})$$

is relatively $\mathcal{I}$ -compact in $\text{AE}(X)$, so ϕ is $\mathcal{I}$ -Lipschitz compact.

By the hypothesis, there exists $f \in \text{Lip}_{0\mathcal{F}}(Z, \text{AE}(X))$ such that

$$\sup_{z \in B_Z} \|\delta_X(\phi(z)) - f(z)\|_{\text{AE}(X)} < \varepsilon.$$

In particular this estimate holds for every $z \in K$. Since d_Z is a constant multiple of the inherited metric from X , f is also Lipschitz on Z with respect to the inherited metric from X . Applying Lemma 1 with target $E = \text{AE}(X)$, extend f to a map $G \in \text{Lip}_{0\mathcal{F}}(X, \text{AE}(X))$. For $x \in K$, $G(x) = f(x)$, and therefore

$$\sup_{x \in K} \|\delta_X(x) - G(x)\|_{\text{AE}(X)} < \varepsilon.$$

This is exactly the closure condition defining the $\mathcal{I}$ -Lipschitz approximation property with respect to $\text{Lip}_{0\mathcal{F}}(X, \text{AE}(X))$. $\square$

Corollary

Together with Proposition 4 of Rueda–Sanchez–Perez, equivalently Proposition 4.9 of Achour–Rueda–Sanchez–Perez–Yahi, Theorem 1 gives the equivalence:

$$\begin{aligned} X \text{ has the } \mathcal{I}\text{-Lipschitz approximation property with respect to } \text{Lip}_{0\mathcal{F}}(X, \text{AE}(X)) \\ \iff \delta_X \circ \phi \text{ is uniformly approximable on } B_Z \\ \text{by finite-rank Lipschitz maps for every} \\ \mathcal{I}\text{-Lipschitz compact } \phi : Z \rightarrow X. \end{aligned}$$

Verification Notes

No computational verification is needed. The proof uses only:

- the source definitions of relatively $\mathcal{I}$ -Lipschitz compact sets and the Lipschitz- $\tau_{\mathcal{I}}$ closure;
- the observation that relatively $\mathcal{I}$ -compact subsets of a Banach space are bounded;
- scalar McShane extension, applied coordinatewise to a finite-dimensional range.

The main reviewer check is whether the source’s phrase “can be approximated by finite rank operators of $\text{Lip}_{0\mathcal{F}}$ uniformly on B_Z ” uses the standard finite-dimensional-range meaning. That is the meaning used in the cited Lipschitz-operator-ideal paper and in the source’s preceding Proposition 3. Under that reading, the extension lemma is immediate.

Novelty Check

Bounded local checks searched the run indexes for arXiv id ‘1609.02707’ and for nearby phrases including ‘Proposition 4.9’, ‘ $\mathcal{I}$ -Lipschitz compact mapping’, ‘ $\text{Lip}_{0\mathcal{F}}$ ’, ‘uniformly on B_Z ’, and ‘Lipschitz approximation property’. No existing run packet for this exact converse was found.

External web searches on 2026-07-18 used the phrases ‘Proposition 4.9 Lipschitz operator ideals converse’, ‘I-Lipschitz compact mapping uniformly on B_Z converse’, and ‘However the converse is still an open problem I-Lipschitz’. The searches returned the original JMAA paper [2], the arXiv source paper [1], and related Lipschitz-free approximation-property literature, but no direct statement of this converse.

Limitations

This packet does not address the older classical problem in Section 1 of the source: whether

$$\mathcal{K}(E; E) = \overline{E' \otimes E}^{\tau_b}$$

for a Banach space E implies that E has the approximation property.

Human-Review Recommendation

Review as a likely-valid full solution to the Proposition 4 converse. The proof is short; the only likely failure mode is a nonstandard interpretation of finite-rank Lipschitz operator in the source or in $\text{Lip}_{0\mathcal{F}}$.

References

- [1] P. Rueda and E. A. Sanchez-Perez, *The approximation property and Lipschitz mappings on Banach spaces*, arXiv:1609.02707, 2016.
- [2] D. Achour, P. Rueda, E. A. Sanchez-Perez, and R. Yahi, *Lipschitz operator ideals and the approximation property*, Journal of Mathematical Analysis and Applications 436 (2016), 217–236.
- [3] E. J. McShane, *Extension of range of functions*, Bulletin of the American Mathematical Society 40 (1934), 837–842.